UNIT - I

INTRODUCTION TO QUANTUM COMPUTATION

1.1 INTRODUCTION

- **Artificial Intelligence** is the study to provide human intelligence artificially to the machines that make decisions according to its own intelligence to perform the specified task.
- **Quantum Computing** is completely based on the quantum theory. Quantum computing is primarily used for complex computations. It makes the computation faster and provides the efficient results.
- **Classical Computer** uses voltages flowing through circuits and gates, which can be controlled and manipulated entirely by classical mechanics.

Classical computer is built from an electrical circuit containing wires and logic gates

- **Quantum Computer** uses laws of quantum mechanics to perform massively parallel computing through superposition, entanglement and decoherence.

  A quantum computer is built from a quantum circuit containing wires and elementary quantum gates to carry around and manipulate the quantum information.

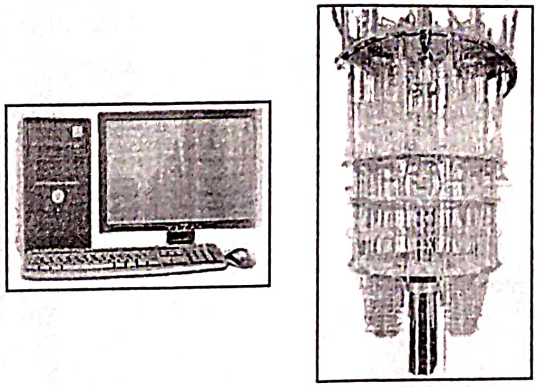


Fig. 1.1: Classical computer Vs Quantum computer

Brief History of Quantum Computers:

- **1959:** Quantum computing tends to trace its roots back to a 1959 speech by Richard P. Feynman in which he spoke about the idea of exploiting quantum effects to create more powerful computers.
- **1981:** Richard Feynman proposed to use quantum computing to model quantum systems. He also describe theoretical model of quantum computer.
- **1985:** David Deutsch described first universal quantum computer
- **1994:** Peter Shor developed the first algorithm for quantum computer (factorization into primes)
- **1995:** Schumacher proposed "Quantum bit" or "qubit" as physical resource
- **1996:** LovGrover developed an algorithm for search in unsorted database
- **1998:** The first quantum computers on two qubits, based on NMR (Oxford; IBM, MIT, Stanford)
- **2000:** Quantum computer on 7 qubits, based on NMR (Los-Alamos)
- **2001:** $15 = 3 \times 5$ on 7-qubit quantum computer by IBM
- **2005-2006:** Experiments with photons; quantum dots; fullerenes and nanotubes as "particle traps"

1.2 OVERVIEW OF QUANTUM COMPUTATION

- Imagine you have a regular computer, it uses bits to store and process information. A bit can be either a 0 or a 1, li light switch that can be either off (0) or on (1).
- Instead of using bits, a quantum computer uses something called quantum bits or qubits. qubits are physical objec like bits, are realized as actual physical systems as shown in Fig. 1.2. Like classical bit a qubit also has a state.
- Two possible states for a qubit are the states (Ket zero)|0>and (Ket one) |1>, which as you mightguess correspond to the states 0 and 1 for a classical bit. Notation like '| >' is called the Dirac notation.

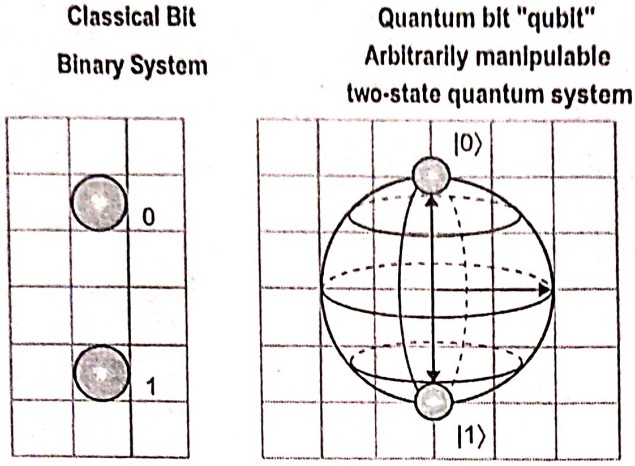


Fig. 1.2: Classical bit and qubit

- Unlike classical bits, qubits can exist in multiple states at the same time. It's a bit like having a light switch that can be both off and on simultaneously! This property is called superposition and it's one of the key features of quantum computing.
- Superposition allows quantum computers to process a vast amount of information all at once, making them incredibly powerful for certain types of problems.

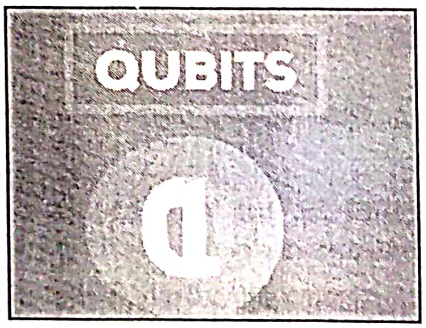


Fig. 1.3: Qubit superposition

Superposition Property:

- ➢ It enables a quantum computer to be in multiple states at once.
- ➢ Superposition is the ability of a quantum system to be in multiple states at the same time until it is measured.
- ➢ The number of possible states grows exponentially as the number of qubits increases.
- ➢ The possible states are represented as 2^n, where n is the number of qubits. number of possible states for five qubi is: $2^5 = 32$ possible states.

- In addition to superposition, qubits can also be entangled with each other. Entanglement means that the state of on qubit is instantly related to the state of another, no matter how far apart they are. It is like having two light switche that are magically connected – flipping one switch will affect the other, even if they are in different rooms!

ntanglement:

- In Quantum Mechanics, it sometimes occurs that a measurement of one particle will effect the state of another particle, even though classically there is no direct interaction. (This is a controversial interpretation).
- When this happens, the state of the two particles is said to be entangled.
- Entanglement is the ability of quantum systems to exhibit correlations between states within a superposition.

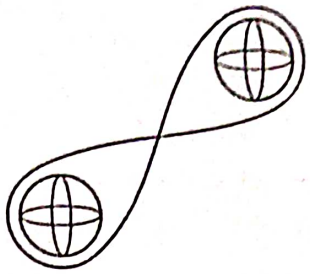


Fig. 1.4: Entanglement

- These unique properties of qubits give quantum computers the ability to perform certain computations much faster than traditional computers, especially for specific tasks like factoring large numbers (useful for cryptography), simulating quantum systems (helpful for drug discovery and material science) and solving complex optimization problems.
- However, building and controlling qubits is incredibly challenging because they are sensitive to their environment and can easily lose their quantum state (a process called decoherence).
- In summary, quantum computing is a fascinating new way of processing information using qubits, which can exist in multiple states at the same time (superposition) and be entangled with each other.
- This enables quantum computers to process information in ways that classical computers cannot, potentially leading to faster and more efficient problem-solving.

1.2.1 Single Qubit Gates

- A quantum bit or a qubit is a fundamental unit of quantum information processing just as a bit is a fundamental unit of classical information processing.
- A single qubit state is represented by a pair of complex numbers $\begin{pmatrix} a \\ b \end{pmatrix}$ where,

$$|a|^2 + |b|^2 = 1.$$

Then, a single qubit state is

$$\begin{pmatrix} a \\ b \end{pmatrix} = a\begin{pmatrix} 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \end{pmatrix} = a|0\rangle + b|1\rangle$$

- So a single qubit can exist in an infinite number of states whereas a bit can exist in either in 0 state or 1 state.
- A single qubit can exist in a superposition of two basis states, conventionally denoted as $|0\rangle$ and $|1\rangle$. These basis states represent the classical binary values 0 and 1.

Now, let's understand single qubit gates and their effects on the qubit states $|0\rangle$ and $|1\rangle$:

- **Identity Gate (I):** The identity gate does not change the state of the qubit. The identity gate, denoted as I. It is represented by the 2×2 identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Applying the identity gate to both $|0\rangle$ and $|1\rangle$ does not change their states.

Mathematically, it can be represented as follows:

$$I|0\rangle = |0\rangle$$

$$I|1\rangle = |1\rangle$$

In other words, applying the I gate to the $|0\rangle$ state keeps it in the $|0\rangle$ state and applying it to the $|1\rangle$ state keeps it in the $|1\rangle$ state.

- **Pauli-X Gate (X):** The Pauli-X gate, denoted as X, is often called a "bit-flip" gate because it flips the state of the qubit. The Pauli-X gate is equivalent to the classical NOT gate. It flips the qubit state. It is represented by the following matrix:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The Pauli-X gate swaps the amplitudes of the $|0\rangle$ and $|1\rangle$ states.

Mathematically, it can be represented as follows:

$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

When you apply the X gate to the $|0\rangle$ state, it changes it to the $|1\rangle$ state and vice versa.

- **Pauli-Y Gate (Y):** The Pauli-Y gate, denoted as Y, is a combination of a bit-flip and a phase change.

The Pauli-Y gate introduces a phase shift and performs a rotation around the y-axis of the Bloch sphere. It is represented by the following matrix:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

The Pauli-Y gate introduces a phase factor and flips the qubit states.

Mathematically, it can be represented as follows:

$$Y|0\rangle = i|1\rangle$$

$$Y|1\rangle = -i|0\rangle$$

When you apply the Y gate to the $|0\rangle$ state, it changes it to the $i|1\rangle$ state (with a phase of i) and applying it to the $|1\rangle$ state changes it to the $-i|0\rangle$ state (with a phase of $-i$).

- **Pauli-Z Gate (Z):** The Pauli-Z gate, denoted as Z, is a phase-flip gate. The Pauli-Z gate introduces a phase shift and performs a rotation around the z-axis of the Bloch sphere. It is represented by the following matrix:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The Pauli-Z gate introduces a phase factor and flips the sign of the $|1\rangle$ state.

Mathematically, it can be represented as follows:

$$Z|0\rangle = |0\rangle$$

$$Z|1\rangle = -|1\rangle$$

When you apply the Z gate to the $|1\rangle$ state, it changes its phase, flipping the sign.

- **Hadamard Gate (H):** The Hadamard gate, denoted as H, creates a superposition between the $|0\rangle$ and $|1\rangle$ states. It is fundamental single-qubit gate in quantum computing. It performs a unique operation on a qubit that creates superposition of the $|0\rangle$ and $|1\rangle$ states. The Hadamard gate is represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Mathematically, it can be represented as follows:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2} = |+\rangle$$

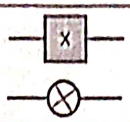
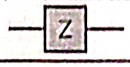
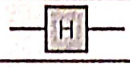
$$H|1\rangle = (|0\rangle - |1\rangle) / \sqrt{2} = |-\rangle$$

Applying the Hadamard gate to the $|0\rangle$ and $|1\rangle$ states produces the plus state $|+\rangle$ and the minus state $|-\rangle$, respectively.

Here, $|+\rangle$ represents a new state called the "plus" state. It is a superposition of the $|0\rangle$ and $|1\rangle$ states with equal probability amplitudes.

Similarly, $|-\rangle$ represents another new state called the "minus" state. It is also a superposition of the $|0\rangle$ and $|1\rangle$ states but with a phase difference.

Table 1.1: Circuit Representations of these Single Qubit Gates

Operator	Gate(s)
Identity Gate (I)	
Pauli-X Gate (X)	
Pauli-X Gate (Y)	
Pauli-X Gate (Z)	
Hadamard Gate (H)	

By combining these gates and applying them in different sequences, quantum computers can perform complex computations and solve problems that would be impractical for classical computers. Single qubit gates are the building blocks of quantum circuits, where multiple qubits can interact and create powerful quantum effects

1.2.2 Multiple Qubit Gates

Multiple-qubit gates also known as multi-qubit gates in quantum computing act on two or more qubits simultaneously, allowing for the entanglement and interaction of multiple qubits.

These gates are represented by larger matrices compared to single-qubit gates. Types of multiple qubit gates:

(i) CNOT (Controlled-NOT) Gate

The CNOT gate is a two-qubit gate that applies a NOT gate (Pauli-X gate) to the target qubit (second qubit) if and only if the control qubit (first qubit) is in the state $|1\rangle$. It flips the target qubit's state when the control qubit is $|1\rangle$, while leaving it unchanged when the control qubit is $|0\rangle$. The CNOT gate is commonly represented as:

$$\text{Controlled NOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

In equations:

$$|00\rangle \rightarrow |00\rangle \; ; \; |01\rangle \rightarrow |01\rangle;$$

$$|10\rangle \rightarrow |11\rangle \; ; \; |11\rangle \rightarrow |10\rangle$$

Any multiple qubit logic gate may be composed from CNOT and single qubit gates.

Mathematically, the CNOT gate is represented as follows:

$$\text{CNOT}|00\rangle = |00\rangle$$

$$\text{CNOT}|01\rangle = |01\rangle$$

$$CNOT|10\rangle = |11\rangle$$

$$CNOT|11\rangle = |10\rangle$$

Let's see an example of the CNOT gate in action:

Suppose we have a two-qubit system in the state $|01\rangle$ and we apply the CNOT gate with the first qubit as the control qu and the second qubit as the target qubit:

$$CNOT(|01\rangle) = CNOT(|0\rangle \otimes |1\rangle) = |01\rangle$$

Since the control qubit is $|0\rangle$, the target qubit remains unchanged.

Now, let's apply the CNOT gate to a two-qubit system in the state $|11\rangle$:

$$CNOT(|11\rangle) = CNOT(|1\rangle \otimes |1\rangle) = |10\rangle$$

The control qubit is $|1\rangle$, so the target qubit undergoes an X operation, changing its state from $|1\rangle$ to $|0\rangle$.

(ii) SWAP Gate

The SWAP gate exchanges the states of two qubits. It swaps the values of two qubits, such that the state of the first qubit becomes the state of the second qubit and vice versa. The SWAP gate is represented as:

$$SWAP = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Mathematically, the SWAP gate is represented as follows:

$$SWAP|00\rangle = |00\rangle$$

$$SWAP|01\rangle = |10\rangle$$

$$SWAP|10\rangle = |01\rangle$$

$$SWAP|11\rangle = |11\rangle$$

Let's see an example of the SWAP gate:

Suppose we have a two-qubit system in the state $|01\rangle$ and we apply the SWAP gate:

$$SWAP(|01\rangle) = SWAP(|0\rangle \otimes |1\rangle) = |10\rangle$$

The states of the two qubits have been swapped. The first qubit is now $|1\rangle$ and the second qubit is $|0\rangle$.

(iii) Controlled-Phase Gate (CZ Gate)

The CZ gate, also known as the controlled-Z gate, applies a phase shift of π radians to the target qubit if and only if the control qubit is in the state $|1\rangle$. It leaves the target qubit unchanged when the control qubit is $|0\rangle$. The CZ gate is represented as:

$$CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Unlike the CNOT gate, the CZ gate does not perform any bit-flip operation but instead manipulates the phase of the target qubit depending on the state of the control qubit.

Mathematically, the CZ gate is represented as follows:

$$CZ|00\rangle = |00\rangle$$

$$CZ|01\rangle = |01\rangle$$

$$CZ|10\rangle = |10\rangle$$

$$CZ|11\rangle = -|11\rangle$$

As you can see, the CZ gate leaves the states $|00\rangle$, $|01\rangle$ and $|10\rangle$ unchanged, while it applies a phase flip (changes the sign) to the state $|11\rangle$.

Let's go through an example of the CZ gate:

Suppose we have a two-qubit system in the state $|01\rangle$ and we apply the CZ gate with the first qubit as the control qubit and the second qubit as the target qubit:

$$CZ(|01\rangle) = CZ(|0\rangle \otimes |1\rangle) = |01\rangle$$

Since the control qubit is $|0\rangle$, the target qubit remains unchanged and no phase shift is applied.

Now, let's apply the CZ gate to a two-qubit system in the state $|11\rangle$:

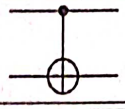
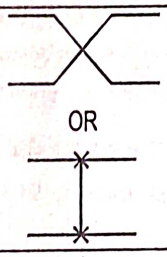
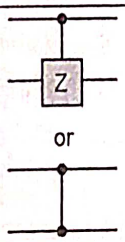
$$CZ(|11\rangle) = CZ(|1\rangle \otimes |1\rangle) = -|11\rangle$$

The control qubit is $|1\rangle$, so the CZ gate applies a phase shift to the target qubit, changing its state from $|11\rangle$ to $-|11\rangle$.

The CZ gate is particularly useful for creating entangled states and performing quantum computations where relative phase information between qubits is essential.

It plays a significant role in quantum algorithms such as quantum error correction and quantum teleportation.

Table 1.2: Circuit Representations of these Single Qubit Gates

Operator	Gate
CNOT	
SWAP	
Controlled-Phase Gate	

Multiple-qubit gates play a crucial role in quantum algorithms, enabling the creation of entanglement and the implementation of complex quantum operations.

They are essential building blocks for quantum circuits and computations.

1.2.3 Measurements in Bases Vs Computational Basis

In quantum computing, measurements can be performed in different bases, including the computational basis.

(a) Computational Basis Measurements

- In the computational basis, measurements are performed in the standard basis states $|0\rangle$ and $|1\rangle$.
- When a qubit is measured in the computational basis, it collapses into one of the basis states with probabilities determined by the amplitudes of the qubit's quantum state.
- For example, if a qubit is in the state $\alpha|0\rangle + \beta|1\rangle$, where α and β are probability amplitudes, the measurement will yield $|0\rangle$with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

(b) Measurements in Other Bases

- In this type of measurement, the quantum computer reads the qubit in the custom basis formed, not just as 0 and 1. The custom basis states $|+\rangle$ and $|-\rangle$ are superposition of the computational basis states $|0\rangle$ and $|1\rangle$.
- When a measurement is made in the custom basis, the quantum state collapses to either $|+\rangle$ or $|-\rangle$ with corresponding probabilities.
- These measurements enable quantum algorithms to manipulate and extract information from quantum states in unique ways, leveraging the power of quantum superposition and entanglement.

1.2.4 Quantum Circuits

A quantum circuit is a visual representation of a series of quantum operations or quantum gates, that manipulate quantum bits or qubits to perform computations in quantum computing. The key component of quantum circuits are qubits, quantum gates, entanglement and measurement.

(i) Components of a Quantum Circuit:

- **Qubits:** Qubits are the basic units of quantum information. Unlike classical bits, which can be in either a 0 or 1 state, qubits can exist in a combination of both 0 and 1, known as a superposition. This unique property enables quantum circuits to perform multiple calculations simultaneously.
- **Quantum Gates:** Quantum gates are the building blocks of quantum circuits. Each gate applies a specific transformation to the qubits, altering their quantum state. Examples of quantum gates include the Hadamard gate, Pauli gates (X, Y, Z), CNOT gate and others. Each gate corresponds to a unitary transformation that evolves the quantum state.
- **Entanglement:** Quantum circuits can create entanglement between qubits. Entanglement is a phenomenon where qubits become strongly correlated, regardless of the spatial distance between them. Entangled qubits exhibit non-classical correlations and manipulating one qubit can instantaneously affect its entangled partner.
- **Measurement:** At the end of the quantum circuit, a measurement is performed on the qubits. This process collapses the superposition of each qubit into a classical state (either 0 or 1) with probabilities determined by the quantum state's amplitudes.
- **Quantum Registers:** Quantum registers are collections of qubits grouped together to perform operations collectively. They allow for the manipulation of multiple qubits simultaneously.

(ii) Computational Process:

- **Initialization:** A quantum circuit starts with qubits initialized in a known state, usually $|0\rangle$ or $|1\rangle$. However, qubits can also be prepared in superposition states by applying quantum gates like the Hadamard gate.
- **Quantum Gates Application:** Quantum gates are applied sequentially to the qubits, transforming their quantum state. This stage is where quantum parallelism comes into play, as multiple gates can operate simultaneously on different qubits. These gates include single-qubit gates (e.g., Pauli gates, Hadamard gate) and multi-qubit gates (e.g., CNOT gate, SWAP gate). Each gate corresponds to a unitary transformation that evolves the quantum state.
- **Entanglement Generation:** Some quantum gates create entanglement between qubits, establishing powerful correlations that can be exploited for certain quantum algorithms.
- **Measurement:** The final step is to measure the qubits which we represent by a 'meter' symbol. The outcome of the measurement is probabilistic due to the superposition nature of qubits and the probabilities of different measurement outcomes are determined by the quantum state's amplitudes.

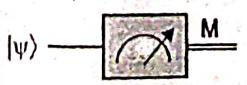


Fig. 1.5: Quantum circuit symbol for measurement

(iii) Representation of Quantum Circuits

- Quantum circuits are typically represented using circuit diagrams. In these diagrams, qubits are represented by horizontal lines and gates are represented by specific symbols corresponding to their operations.
- The direction of time is from left to right, indicating the flow of computation.
- Quantum circuits can be composed of multiple layers or levels, where gates in the same layer are applied simultaneously and information can flow from one layer to the next.
- This layer-based structure allows for parallel and sequential execution of quantum operations.

(iv) Applications:

Quantum circuits are at the heart of quantum computing and are used to design quantum algorithms to solve specific problems efficiently. Quantum circuits can outperform classical algorithms in areas like factoring large numbers (Shor's algorithm), simulating quantum systems (quantum chemistry) and optimization tasks (Grover's algorithm).

1.2.5 Qubit Copying Circuit

- In classical computing, you can easily copy information from one bit to another using standard logical operations like AND, OR or NOT gates.
- However, in the quantum world, copying a quantum state from one qubit to another is not straightforward due to a fundamental principle in quantum mechanics known as the "no-cloning theorem."
- However, there is a concept known as "quantum state teleportation" that allows for the transfer of quantum information from one qubit to another.
- Quantum teleportation is a quantum communication protocol that allows the transfer of the exact quantum state of one qubit onto another qubit, but it requires the classical transmission of information as well.
- The basic idea behind quantum teleportation involves two parties: the sender (Mahesh) and the receiver (Ramesh), who share an entangled pair of qubits known as a Bell pair. Here is a high-level description of the quantum teleportation circuit:
  - ➢ Mahesh and Ramesh share an entangled Bell pair of qubits, where each qubit is in a superposition of states: $|00\rangle + |11\rangle$.
  - ➢ Mahesh wants to send his qubit (let's call it qubit A) to Ramesh.
  - ➢ Mahesh applies a controlled NOT (CNOT) gate to his qubit A, controlled by the state of the qubit he wants to send (let's call it qubit C) and then applies a Hadamard gate to qubit C. This operation is known as a Bell measurement.
  - ➢ Mahesh measures the states of qubits A and C.
  - ➢ Mahesh sends the measurement results (two classical bits) to Ramesh through a classical communication channel.
  - ➢ Based on the measurement results, Ramesh performs certain operations on his qubit (let's call it qubit B) to transform it into the desired state that was originally held by Mahesh qubit A.
- It's important to emphasize that quantum teleportation does not involve copying the quantum state itself but rather transferring the state to another qubit through the entanglement of qubits.

1.3 EXAMPLE: BELL STATES AND QUANTUM TELEPORTATION

1.3.1 Bell States

- Bell states, also known maximally entangled states means their quantum states are correlated in a highly non-classical way.
- There are four Bell states, each represented by a combination of the states of two qubits, usually denoted as $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\psi^+\rangle$ and $|\psi^-\rangle$. The general form of a Bell state is:

$$|\Phi^+\rangle = 1/\sqrt{2}\,(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = 1/\sqrt{2}\,(|00\rangle - |11\rangle)$$

$$|\psi^+\rangle = 1/\sqrt{2}\,(|01\rangle + |10\rangle)$$

$$|\psi^-\rangle = 1/\sqrt{2}\,(|01\rangle - |10\rangle)$$

Here, $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$ represent the basis states of two qubits in the computational basis.

- The coefficients $1/\sqrt{2}$ ensure that the total probability of the system remains normalized.
- Bell states are typically generated using quantum gates and operations. For example, applying a Hadamard gate to the first qubit and a controlled-NOT gate (CNOT) with the first qubit as the control and the second qubit as the target can create the Bell state $|\Phi^+\rangle$.

1.3.2 Quantum Teleportation

Quantum teleportation is a quantum communication protocol that allows the transfer of an unknown quantum state from one qubit to another, without physically transporting the qubit itself.

It relies on the principles of entanglement and quantum measurement. The basic idea behind quantum teleportation is as follows:

- **Initialization:** Mahesh and Ramesh, the sender and receiver, respectively, share an entangled pair of qubits called a Bell pair. The Bell pair is typically in one of the four Bell states: $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\psi^+\rangle$ or $|\psi^-\rangle$.
- **State Preparation:** Mahesh possesses the qubit (let's call it qubit A) that she wants to teleport to Ramesh, while Ramesh has an empty qubit (let's call it qubit B) that will receive the teleported state.
- **Bell Measurement:** Mahesh performs a joint measurement, known as a Bell measurement, on her qubit A and the entangled qubit of the Bell pair.
- **Measurement Results:** Mahesh obtains two classical bits as the measurement outcomes, which convey information about the state of her qubit A.
- **Classical Communication:** Mahesh sends the two classical bits to Ramesh through a classical communication channel.
- **Ramesh's Operations:** Based on the received measurement results, Ramesh performs a specific set of operations (quantum gates) on his qubit B to transform it into the exact replica of Mahesh's original qubit A.

1. **Initial State:**

   Maheshqubit (qubit A): $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex probability amplitudes.

   Rameshqubit (qubit B): Initially, qubit B and Mahesh's qubit A are not entangled.

2. **Entanglement Generation:**

   Before the teleportation process begins, Mahesh and Ramesh share an entangled pair of qubits B and C. This entangled state is known as a Bell pair.

   The Bell pair shared by Mahesh and Ramesh:

$$|\phi\rangle = (1/\sqrt{2}) * (|00\rangle + |11\rangle)$$

3. **Teleportation Process:**

- Mahesh applies a Bell measurement (a specific measurement) on her qubits A and B.
  - ➢ The Bell measurement consists of two operations: a Hadamard gate (H) and a Controlled-NOT gate (CNOT).
  - ➢ The Hadamard gate (H) is applied to qubit A.
  - ➢ The CNOT gate is applied with qubit A as the control qubit and qubit B as the target qubit.
  - ➢ The combined operation on qubits A and B is represented as:

$$(H \otimes I) * CNOT$$

- Entangled state of qubits A and B and Mahesh obtains two classical bits (b_1, b_2) as an outcome.
- Mahesh sends the two classical bits (b_1, b_2) to Ramesh through classical communication.

4. Ramesh's Action:

- Based on the received classical bits (b_1, b_2), Ramesh performs one of four possible quantum operations (X, Z, ZX or I) on his qubit B (the one he shares with Mahesh).
- The operation Ramesh performs on qubit B depends on the classical bit values as follows:
  - ➢ $b_1 = 0$ and $b_2 = 0$: Do nothing (Identity operation – I)
  - ➢ $b_1 = 0$ and $b_2 = 1$: Apply Pauli-Z gate (Z)
  - ➢ $b_1 = 1$ and $b_2 = 0$: Apply Pauli-X gate (X)
  - ➢ $b_1 = 1$ and $b_2 = 1$: Apply Pauli-X gate (X) followed by Pauli-Z gate (Z) (ZX)

5. Result:

- After Ramesh performs the quantum operation on qubit B, qubit B now holds the exact quantum state that Mahesh initially had on her qubit A. The teleportation process is complete.
- Through this protocol, the quantum state of qubit A is faithfully transmitted to qubit B in Ramesh's possession. It's important to note that the actual qubit A is not physically transmitted or cloned in the process. Instead, the entangled Bell pair and the classical communication of measurement results enable the transfer of the quantum information.

1.4 BASICS OF LINEAR ALGEBRA

- A vector is like a list of numbers and it has a specific size, represented by N. So, a vector of dimension N looks like this:

$$[a_0, a_1, \ldots, a_{N-1}]$$

- Normally, we use subscripts like a_k to refer to the k-th element in the vector (e.g., a_0 means the first element).

  But instead, we will use a(k) to represent the same thing (e.g., a(0) means the first element). It may look different, but a(k) is the same as a_k.
- The reason for using a(k) is that sometimes vectors can be better understood when indexed by other sets, not just the usual 0, 1, 2, ..., N–1. This notation gives us more flexibility in understanding vectors in different contexts.
- An example of a vector of dimension 4 would be written as a(0), a(1), a(2), a(3) for its elements. Alternatively, we can use a(00), a(01), a(10), a(11), using Boolean strings to index the elements.
- Using a functional notation like a(k) implies that vectors are like pieces of code that perform computations. For example, a standard basis vector e_k is a piece of code that has a value of 1 at the k-th position and 0 everywhere else.
- When $N = 2^n$, we index the complex coordinates from 0 to N–1 and $\{0, 1\}^n$ as $x_0, x_1, \ldots, x_{N-1}$, but we index the binary string places from 1 to n. This helps differentiate them when considering them as pieces of code.
- So, the functional notation allows us to think of vectors as computational entities and it fits well with some mathematical concepts related to functions and sets.

1.4.1 Hilbert Spaces

- Hilbert space is a vector space over the complex numbers (or sometimes over the real numbers) equipped with an inner product.
- An inner product is a mathematical operation that takes two vectors in the space and produces a complex number (or a real number) as its result.
- We will use H_N to denote this space of dimension N. The elements are real vectors x of dimension N. They are added and multiplied by scalars in the usual manner:
- If x, y are vectors in this space, then so is x + y, which is defined by

$$\begin{bmatrix} x(0) \\ \vdots \\ x(N-1) \end{bmatrix} + \begin{bmatrix} y(0) \\ \vdots \\ y(N-1) \end{bmatrix} = \begin{bmatrix} x(0) + y(0) \\ \vdots \\ x(N-1) + y(N-1) \end{bmatrix}$$

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- The abstract essence of a Hilbert space is that each vector has a norm: The norm of a vector x, really just defined to be

$$||x|| = \left| \sum_k x(k)^2 \right|^{1/2}$$

- A Hilbert space simply generalizes x to many dimensions.
- A unit vector is just a vector of norm 1. Unit vectors together comprise the unit sphere in any Hilbert space.
- Hilbert spaces are fundamental mathematical structures in quantum mechanics that provide a mathematical fra for representing quantum states and performing quantum operations.
- In the context of quantum computing, Hilbert spaces are used to describe the state space of qubits and the evo of quantum systems.
- A Hilbert space is a complex vector space with an inner product and it is complete with respect to the norm induce the inner product.
- In simpler terms, it is a vector space where you can add vectors and multiply them by scalars and it allows for defin an inner product that measures the "angle" between vectors.
- Completeness means that any Cauchy sequence (a sequence that gets arbitrarily close to a limit) of vectors in th Hilbert space converges to a vector within the same space.
- In quantum computing, a single qubit is represented by a two-dimensional Hilbert space, usually denoted as C (complex 2–dimensional space). The basis states $|0\rangle$ and $|1\rangle$ of a single qubit form an orthonormal basis for this Hilbe space:

$$|0\rangle = [1, 0]^T \; |1\rangle = [0, 1]^T$$

- The inner product of two qubit states $|\psi\rangle$ and $|\phi\rangle$ is defined as $\langle\psi|\phi\rangle$ and it satisfies certain properties like linearit conjugate symmetry and positive definiteness.
- Here's an example of a two-qubit Hilbert space ($C^2 \otimes C^2$) where the basis states are represented as tensor products individual qubit basis states:

$|00\rangle = |0\rangle \otimes |0\rangle = [1, 0, 0, 0]^T \; |01\rangle = |0\rangle \otimes |1\rangle = [0, 1, 0, 0]^T \; |10\rangle = |1\rangle \otimes |0\rangle = [0, 0, 1, 0]^T \; |11\rangle = |1\rangle \otimes |1\rangle = [0, 0, 0, 1]^T$

- These four states form an orthonormal basis for the two-qubit Hilbert space. The inner product in this space allows to calculate probabilities and amplitudes for various quantum states.
- Hilbert spaces provide the mathematical framework that enables us to represent quantum states, design quantu algorithms and understand quantum phenomena. They are essential for working with quantum mechanics a quantum computing.

1.4.2 Products and Tensor Products

- There are two ways to combine two Hilbert spaces, m-dimensional H_1 and n-dimensional H_2:
  - (i) The Cartesian product
  - (ii) The Tensor product.

(i) Cartesian Product: The Cartesian product of these spaces obtained by concatenating vectors from the first space w the second space. The resulting space will have (m + n) dimensions and its vectors will look like a(1), a(2), ... , a(m + n)

(ii) Tensor Product: Their tensor product $\mathbb{H}\ 1 \otimes \mathbb{H}\ 2$, is different. If the first space has m dimensions and the second spa has n dimensions, the tensor product will create a space with (m × n) dimensions. The vectors in this space will be the form a(k), where $1 \le k \le m \times n$. Each k corresponds to a pair of indices (i, j) where i ranges from 1 to m and j rang from 1 to n. So, we can write them as a(ij).

- The tensor product of two vectors a and b is defined as $c = a \otimes b$ and its elements are given by $c(ij) = a(i) \times b(i)$.

- In quantum mechanics, a vector representing a pure quantum state is called "separable" if it can be expressed as the tensor product of two other vectors. If not, it is called "entangled".
- In quantum algorithms, we often work with the product of two Hilbert spaces, where each space uses binary strings as indices. The resulting vectors have the form a(xy), where x ranges over the indices of the first space and y over the indices of the second space. However, writing a(xy) does not necessarily mean that a is separable. It could be entangled as well.

Quantum States and Products

- In quantum mechanics, a quantum state represents the state of a quantum system and it is described using a mathematical object called a state vector. The state vector is a complex vector in a vector space known as the Hilbert space, which represents all possible states of the quantum system.
- For a simple example, consider a two-level quantum system, often referred to as a qubit. The state of this qubit can be represented by a two-dimensional complex vector:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- Here, $|0\rangle$ and $|1\rangle$ are the basis states of the qubit, representing the two possible states of the system. The coefficients α and β are complex numbers and their magnitudes squared ($|\alpha|^2$ and $|\beta|^2$) give the probabilities of measuring the qubit in the $|0\rangle$ or $|1\rangle$ state, respectively. The state vector must be normalized, meaning that $|\alpha|^2 + |\beta|^2 = 1$.
- A state vector is often depicted as a point on the surface of a 3-dimensional sphere, known as the Bloch sphere, as shown in Fig. 1.6.
- Two real parameters θ and ψ are sufficient to describe a state vector, since state vectors are constrained to have norm 1 and are equivalent up to global phase. Points on the surface of the Bloch sphere can also be expressed in Cartesian coordinates as

$$(x, y, z) = (\sin\theta \cos\psi, \sin\theta \sin\psi, \cos\theta).$$

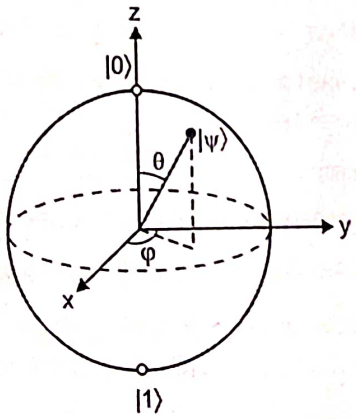

Fig. 1.6: Bloch sphere

Tensor Products

- When dealing with composite quantum systems, like multiple qubits, we use tensor products to represent the combined state of the system. The tensor product combines the state vectors of individual quantum systems into a single state vector representing the joint state of the composite system.
- The tensor product of two quantum states $|a\rangle$ and $|b\rangle$ is denoted by $|a\rangle \otimes |b\rangle$ and is calculated as follows:

$$|a\rangle \otimes |b\rangle = |a\rangle \otimes (\beta|0\rangle + \gamma|1\rangle)$$

$$= \alpha|0\rangle \otimes |0\rangle + \alpha|0\rangle \otimes |1\rangle + \beta|1\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle$$

- So, the tensor product of two qubits results in a composite state vector that represents a four-dimensional Hilbert space. In general, if we have n qubits, the combined state space will be 2^n dimensional.

- **Tensor Product of Operators:** In addition to state vectors, we also use operators to describe quantum ... such as quantum gates. When we apply operations to composite systems, we use the tensor product of oper...

  For example, if we have two single-qubit quantum gates represented by matrices A and B, the tensor produ... gates (A ⊗ B) represents their joint operation on a composite system of two qubits.
- The tensor product is a fundamental concept in quantum computing as it allows us to understand and m... multi-qubit systems, which form the basis for more complex quantum algorithms and computations.
- Overall, products and tensor products are essential tools for representing and understanding quantum state... operations in quantum computing. They provide a mathematical framework for working with composite qu... systems, allowing us to analyze and control quantum information more effectively.

1.4.3 Matrices

- Matrices represent linear operators on Hilbert spaces.
- The matrix I_N denotes the N × N identity matrix. We use square brackets for matrix entries to distinguish them fur... from vector amplitudes, so that the identity matrix has entries $I_N[r, c] = 1$ if $r = c$, $I_N[r, c] = 0$ otherwise. One of the ... properties of matrices is that they define linear operations, namely:

$$U(a + b) = Ua + Ub.$$

  If U is a matrix, then we use U^k to denote the k-th power of the matrix,

$$U^k = UU \ldots U^k \text{ copies}$$

- The transpose of a matrix U is obtained by flipping its rows and columns. We use UT to denote the transpose o... matrix.
- The inner product of two real vectors a and b is a way to measure their similarity.

  It is calculated by multiplying the corresponding elements of the vectors and then summing them up.
- A real matrix U is called unitary if its transpose multiplied by itself gives the identity matrix I.
- Unitary matrices are crucial in quantum computing because they preserve the length of vectors, meaning they do ... distort distances or angles between vectors.

Example of 2 X 2 Unitary Matrix are:

- The identity matrix I, which does not change the vector it operates on.
- The X matrix, also known as the Pauli-X gate, which flips the components of the vector.
- The Hadamard matrix H, which requires a scaling factor to preserve vector length but performs a uniq... transformation involving addition and subtraction of vector components.

(a) Complex Spaces and Inner Products

- A complex matrix U is unitary provided U*U = I. Furthermore, the inner product of two complex vectors a and b ... defined to be

$$\langle a, b \rangle = \sum_{k=0}^{m} \overline{a(k)}\, b(k)$$

- In complex spaces deal with complex numbers, which have both real and imaginary parts. The imaginary unit ... denoted by "i" (i.e., $\sqrt{-1}$). The conjugate of a complex number "z" (written as z*) is obtained by changing the sign ... its imaginary part. For example, if $z = x + iy$, then $z^* = x - iy$.
- The adjoint of a matrix "U" in complex spaces, denoted as "U*," is obtained by taking the conjugate of its transpos... The entries of the vectors and matrices remain the same as in the real case, but when dealing with complex entries, ... need to conjugate them.

- A complex matrix "U" is unitary if its adjoint multiplied by itself gives the identity matrix I. The inner product of two complex vectors "a" and "b" is defined similarly to the real case, but now it involves taking the conjugate of the entries and then performing the sum.
- The concepts of tensor products of matrices remain the same for complex spaces.
- An inner product $\langle v_2|v_1\rangle$ or dot product, on a complex vector space V is a complex function defined on pairs of vectors $|v_1\rangle$ and $|v_2\rangle$, satisfying

$\langle v|v\rangle$ is non-negative real,

$$\langle v_2|v_1\rangle = \overline{\langle v_1|v_2\rangle} \text{ , and}$$

$$(a\langle v_2| + b\langle v_3|)|v_1\rangle = a\langle v_2|v_1\rangle + b\langle v_3|v_1\rangle$$

where $\overline{z}$ is the complex conjugate $\overline{z} = a - lb$ of $z = a + lb$.

If $$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \text{ and } |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix},$$ the inner product

$$\langle a|b\rangle = \langle a||b\rangle$$

$$= (\overline{a}_1\ \overline{a}_2 \dots \overline{a}_n) \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \sum_{i=1}^{n} \overline{a}_i\, b_i$$

1.4.4 Matrices, Graphs and Sums Over Paths

- One rich source of real-valued matrices is graphs.
- A graph G consists of a set V of vertices, also called nodes, together with a binary relation E on V whose members are called edges.
- The adjacency matrix A = A(G) of a graph G = (V,E) is defined for all u, v ∈ V by:

$$A[u, v] = \begin{cases} 1 & \text{if } (u, v) \in E \\ 0 & \text{Otherwise} \end{cases}$$

If E is a symmetric relation, then A is a symmetric matrix. In that case, G is an undirected graph; otherwise it is directed.

- The degree of a vertex in a graph is the number of edges connected to that vertex.
- If every vertex in a graph has the same degree, which is d, then the graph is called "regular". consider the matrix A' = (1/d) A.
- For a regular graph with degree d, we create a matrix called the stochastic adjacency matrix, denoted as A, where every entry is divided by d. This matrix ensures that every row and column sums to 1.
- If a regular graph is also bipartite, which means we can divide its vertices into two sets such that all edges connect vertices from different sets, then we can further modify the matrix to create a unitary matrix.
- A unitary matrix is a special kind of matrix that preserves the length of vectors.
- It is characterized by having all its rows and columns with length 1 and the dot product between any two distinct rows or columns is 0 (i.e., they are orthogonal). To convert the stochastic adjacency matrix to a unitary matrix, we need to adjust the entries in a specific way.
- The example given is for a graph called the "four-cycle," where the unitary matrix is obtained by multiplying the stochastic adjacency matrix by a scaling factor and making some specific changes to certain entries.
- This unitary matrix satisfies the properties of a unitary matrix, making it useful in certain quantum algorithms.

CASE STUDY: TO CREATE A QUANTUM GATE FROM A UNITARY MATRIX IN QISKIT

https://quantumcomputinguk.org/tutorials/how-to-create-a-quantum-gate-from-a-unitary-matrix-in-qiskit-wit

You Can Follow These Steps:

Step 1: Import the necessary libraries and set up the quantum circuit.

```
fromqiskit import QuantumRegister, ClassicalRegister
fromqiskit import QuantumCir uit, execute
fromqiskit.quantum_info.oper ators import Operator
fromqiskit import Aer
# Create a quantum circuit with the required number of qubitsnum_qubits = 2 qc = QuantumCircuit(num_qu
```

Step 2: Define the unitary matrix you want to use as the gate.

```
    Importnumpy as np
# Define the unitary matrix (example: Hadamard gate) unitary_matrix = np.array([[1, 1], [1, - 1]]) / np.sqrt(2
```

Step 3: Apply the unitary matrix as a gate to the quantum circuit.

```
  # Apply the unitary matrix as a gate to the circuit
      qc.unitary(unitary_matrix, [0, 1])
# Apply the gate to qubits 0 and 1
```

Step 4: Visualize the quantum circuit.

```
  print(qc)
```

Step 5: Simulate and execute the quantum circuit to obtain the results.

```
  # Simulate the quantum circuit
  simulator = Aer.get_backend('statevector_simulator')
  job = execute(qc, simulator)
  result = job.result()
  # Get the final state vector
  statevector = result.get_statevector()
  print("Final state vector:")
  print(statevector)
```

EXERCISE

Quantum Computation:

1. What is a qubit and how does it different from a classical bit?
2. Differentiate the Quantum computer with classical computer
3. What are single qubit gates in quantum computation and how do they operate on qubits?

4. How do multiple qubit gates different from single qubit gates and what are some common examples of multiple qubit gates?
5. What is the difference between measurements in bases and measurements in the computational basis in quantum computing?
6. Can you explain the concept of a quantum circuit and how it is used to represent quantum algorithms?
7. What are Bell states and why are they important in quantum information processing?
8. Explain the concept of quantum teleportation and how it utilizes entanglement to transfer quantum information between qubits.
9. Explain the concept of superposition in quantum computation with respect to qubits.
10. Name and describe the three fundamental single-qubit gates commonly used in quantum computing.
11. How do multiple-qubit gates, such as the Controlled-NOT (CNOT) gate, operate in quantum circuits?
12. Describe the difference between measurements in bases other than the computational basis and measurements in the computational basis.
13. What are quantum circuits and how are they used to manipulate qubits?
14. Can a quantum circuit copy the state of a qubit like classical circuits? Why or why not?
15. Provide an example of a quantum circuit that creates a Bell state.
16. Explain the concept of quantum teleportation and how it utilizes entanglement.
17. How is quantum entanglement utilized in quantum teleportation?

Basics of Linear Algebra:

18. Define Hilbert spaces and explain their significance in quantum mechanics.
19. What is the tensor product of two vector spaces and how is it used in quantum mechanics?
20. Describe how to perform matrix multiplication and its importance in linear algebra.
21. Explain the concept of graphs in the context of linear algebra.
22. How are graphs represented mathematically in linear algebra?
23. What are paths in a graph and how are they related to linear algebra?
24. Describe the concept of a sum over paths in the context of quantum mechanics.
25. Provide an example of a sum over paths calculation in a quantum system.
26. How does linear algebra play a crucial role in the manipulation of quantum states in quantum computing?
27. What are some practical applications of quantum computing that rely heavily on linear algebra?
28. How does linear algebra play a fundamental role in understanding quantum computations and representations?
29. What are Hilbert spaces and how are they used to describe quantum systems?
30. How do matrix products and tensor products relate to quantum computations and what are their applications in quantum algorithms and quantum states?
31. Explain Measurements in bases Vs computational basis.

32. Consider the matrix $\begin{bmatrix} w & w \\ w & -w \end{bmatrix}$

For what real values of w is it a unitary matrix?

33. Consider the matrix U equal to:

$$\frac{1}{\sqrt{2}}\begin{bmatrix} I_N & I_N \\ I_N & -I_N \end{bmatrix}$$

Show that for any N, the matrix U is unitary.

34. Explain the four Pauli matrices, I, X, Y and Z, form an orthonormal basis for the space of 2 × 2 matrices, regarded as 4-dimensional complex Hilbert space.

✠ ✠ ✠

UNIT - II

QUANTUM ARCHITECTURE

2.1 INTRODUCTION

Quantum architecture refers to the physical design and layout of quantum computing systems, including the hardware components and their interconnections. Quantum computing aims to harness the principles of quantum mechanics to perform computations that are beyond the capabilities of classical computers.

Quantum Architecture

Quantum architecture refers to the physical design and layout of quantum computing systems, including the hardware components and their interconnections. Quantum computing aims to harness the principles of quantum mechanics to perform computations that are beyond the capabilities of classical computers.

Unlike classical computers that use bits to represent information as either 0 or 1, quantum computers use quantum bits or qubits, which can represent 0, 1 or both states simultaneously, a phenomenon known as superposition. This property of superposition allows quantum computers to perform certain calculations exponentially faster than classical computers for specific tasks.

As shown in Fig. 2.1, Quantum architecture involves several key components:

- **Qubits:** These are the basic building blocks of quantum computing and are the quantum analogs of classical bits. Qubits are typically implemented using physical systems such as superconducting circuits, trapped ions or topological qubits.
- **Quantum Gates:** Quantum gates are the quantum counterparts of classical logic gates. They manipulate qubits, enabling quantum computers to perform quantum algorithms. Common quantum gates include Hadamard gate, CNOT (Controlled-NOT) gate and Toffoli gate, among others.
- **Quantum Registers:** A quantum register is a group of qubits that can be manipulated together to perform specific quantum operations.
- **Quantum Circuits:** Quantum circuits are sequences of quantum gates applied to qubits in a specific order to execute quantum algorithms. They represent the quantum version of classical computer programs.
- **Quantum Interconnects:** Quantum interconnects are the physical connections between qubits that allow for communication and entanglement, a crucial property in quantum computing that enables quantum parallelism and enhanced computational power.
- **Error Correction:** Quantum computations are susceptible to errors caused by environmental noise and decoherence. Quantum error correction techniques are used to protect quantum information from such errors and ensure reliable computations.
- **Cooling and Isolation:** Quantum computers need to be operated at extremely low temperatures to maintain the delicate quantum states of qubits and reduce interference from the environment.
- **Control Systems:** Control systems are responsible for applying the correct sequences of quantum gates and measurements to the qubits.

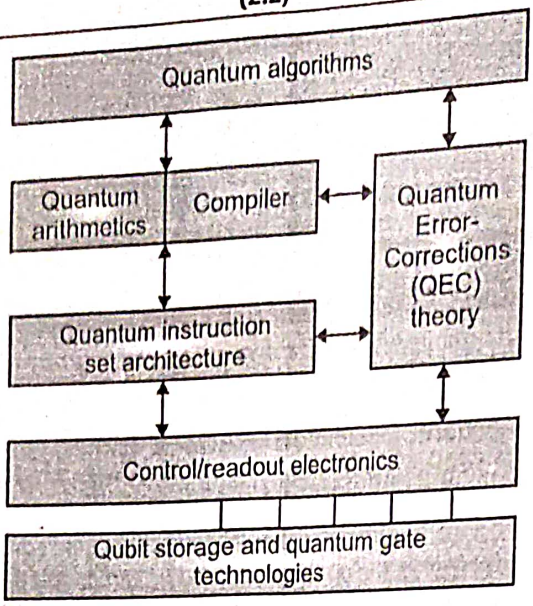

Fig. 2.1: General architecture of a quantum computer, going from a high-level description of a quantum algorithm, to the actual physical operation of the qubits in a layered architecture

2.2 THE FRAMEWORK OF QUANTUM MECHANICS

Quantum mechanics is a fundamental framework of physics that describes the behavior of particles and systems at the atomic and subatomic scales. It is a cornerstone of modern physics and has been highly successful in explaining a wide range of phenomena that classical mechanics and classical electromagnetism cannot account for.

The framework of quantum mechanics is built on several key principles and mathematical formalisms:

- **Wave-Particle Duality:** One of the most intriguing aspects of quantum mechanics is the wave-particle duality, which suggests that particles, such as electrons and photons, exhibit both wave-like and particle-like properties. This means that these particles can behave as discrete entities (particles) and spread out as waves, depending on the experimental setup.
- **Superposition:** Quantum superposition is the principle that a quantum system can exist in multiple states simultaneously. For example, a qubit in a quantum computer can be in a state representing both 0 and 1 at the same time.
- **Quantum States and Wave Functions:** The quantum state of a particle or a system is described by a mathematical entity called a wave function (also known as a state vector). The wave function encodes the probability amplitude for finding the particle in various states upon measurement.
- **Uncertainty Principle:** The Heisenberg uncertainty principle is a fundamental concept in quantum mechanics, stating that certain pairs of physical properties, such as position and momentum, cannot be simultaneously measured with arbitrary precision. This introduces inherent uncertainty into quantum systems.
- **Measurement and Collapse:** When a quantum system is measured, the wave function "collapses" into one of its allowed states, with the probability of each state determined by the square of the amplitude of its corresponding wave function component. This process is inherently probabilistic.
- **Entanglement:** Entanglement is a unique quantum phenomenon where two or more particles become correlated in such a way that the state of one particle is instantly related to the state of the others, regardless of the distance between them. This property is at the heart of quantum information processing and quantum teleportation.
- **Operators and Observables:** In quantum mechanics, physical observables, such as position, momentum and spin, are associated with mathematical operators. These operators act on the wave function, allowing us to calculate the expected values of observables in a quantum system.
- **Schrödinger Equation:** The Schrödinger equation is a central equation in quantum mechanics that describes how the wave function of a quantum system evolves over time. It links the energy of the system with the time evolution of its wave function.

The mathematics of quantum mechanics is primarily based on linear algebra and Hilbert space, providing a formal and powerful language to describe and predict the behavior of quantum systems.

Quantum mechanics has wide-ranging applications, including in quantum computing, quantum cryptography, condensed matter physics and particle physics, among others. Despite its tremendous success, quantum mechanics still presents philosophical questions and challenges our intuitive understanding of reality at the quantum level.

2.2.1 The State of a Quantum System

- The state of a quantum system is a fundamental concept in quantum mechanics. Unlike classical systems, where the state can be precisely described by definite values of properties like position and momentum, quantum systems have inherent uncertainty due to the principles of superposition and entanglement.
- The state of a quantum system is mathematically represented by a ket vector, usually denoted as $|\psi\rangle$. This ket vector belongs to a Hilbert space, which is a complex vector space associated with the quantum system.
- The state vector $|\psi\rangle$ contains all the information about the probabilities of different outcomes when making measurements on the quantum system. The probability of observing a specific measurement outcome corresponds to the squared magnitude of the corresponding coefficient of the state vector.
- The state vector $|\psi\rangle$ can be expressed as a superposition of basis states, which are the eigenstates of the observable being measured. For example, in a two-level quantum system (qubit), the basis states are usually denoted as $|0\rangle$ and $|1\rangle$. The state vector $|\psi\rangle$ can then be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex numbers called probability amplitudes and their squared magnitudes give the probabilities of observing the corresponding outcomes, means that satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$.

- Upon measurement, the quantum state "collapses" to one of its possible states with a probability given by the squared magnitude of the corresponding probability amplitude. For the example above, the probability of measuring the qubit in state $|0\rangle$ is $|\alpha|^2$ and the probability of measuring it in state $|1\rangle$ is $|\beta|^2$.
- The state of a quantum system can evolve over time and this evolution is described by the Schrödinger equation. The Schrödinger equation gives the time-dependence of the state vector and describes how the state of the quantum system changes as it evolves over time.
- The state of a qubit can be described by a complex vector in a two-dimensional Hilbert space. The Bloch sphere as shown in Fig. 2.2 provides a unique representation of this quantum state in terms of two angles: θ (theta) and φ (phi).

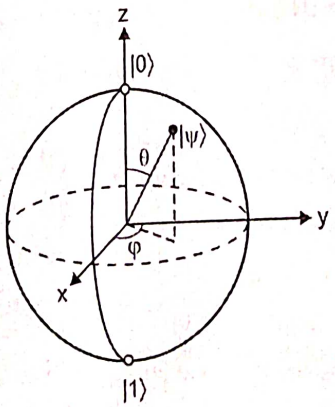

Fig. 2.2: Bloch sphere

- Given a qubit state $|\psi\rangle$, the angles θ and φ can be calculated as follows:
  - ➢ θ is the angle between the vector $|\psi\rangle$ and the positive z-axis (the vertical axis pointing from the center to the north pole of the sphere).
  - ➢ φ is the angle between the projection of $|\psi\rangle$ onto the xy-plane (the equator of the sphere) and the positive x-axis (the horizontal axis pointing to the right).

Mathematically, the state $|\psi\rangle$ can be written as:

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{(i\varphi)}\sin(\theta/2)\,|1\rangle$$

- The Bloch sphere allows us to visualize the quantum state of the qubit as a point on the surface of the sphere.

For example:

- ➢ When $\theta = 0$, the state $|\psi\rangle$ is $|0\rangle$ and the corresponding point is at the center of the sphere.
- ➢ When $\theta = \pi$, the state $|\psi\rangle$ is $|1\rangle$ and the corresponding point is at the north pole of the sphere.
- ➢ For any other values of θ and φ, the point on the sphere represents a superposition of $|0\rangle$ and $|1\rangle$.

- It's important to note that the state of a quantum system evolves deterministically under unitary transformations, but the outcome of a measurement is probabilistic due to the nature of quantum measurement. This feature is what makes quantum systems so intriguing and different from classical systems and it is the basis for their potential for quantum parallelism and exponential speedup in certain computations.

2.2.2 Time-Evolution of a Closed System

The time-evolution of a closed quantum system is a fundamental concept in quantum mechanics that describes how the state of the system changes over time when it is isolated from its external environment and not subject to any external influences or measurements. In other words, a closed system does not exchange energy, information or particles with its surroundings.

Evolution Postulate

- The time-evolution of the state of a closed quantum system is described by unitary operator. That is, for any evolution of the closed system there exists a unitary operator U such that if the initial state of the system is $|\psi_1\rangle$, then after the evolution the state of the system will be

$$|\psi_2\rangle = U|\psi_1\rangle$$

- The time-evolution of a closed quantum system is governed by a unitary operator, often denoted by U(t), where "t" represents time.
- The unitary operator U(t) is a linear transformation that maps the state of the system at an initial time t_0 (represented by a state vector $|\psi(t_0)\rangle$) to its state at a later time t (represented by a state vector $|\psi(t)\rangle$).

Mathematically, the time-evolution of a closed quantum system is described by the Schrödinger equation:

$$i\hbar\,\partial/\partial t\,|\psi(t)\rangle = H\,|\psi(t)\rangle$$

where:

i is the imaginary unit $\left(\sqrt{(-1)}\right)$,

$\hbar$ (h-bar) is the reduced Planck's constant (h divided by 2π),

$\partial/\partial t$ is the partial derivative with respect to time,

$|\psi(t)\rangle$ is the state vector of the system at time t,

H is the Hamiltonian operator of the system.

- The Hamiltonian operator (H) represents the total energy of the system and encodes the system's dynamics. It is a Hermitian operator, meaning that it is equal to its own adjoint (conjugate transpose).
- Solving the Schrödinger equation yields the time-dependent state vector $|\psi(t)\rangle$, which fully describes the state of the closed quantum system at any time t. This state vector contains all the information about the system's quantum properties, such as position, momentum and other observables.
- It is important to note that the time-evolution described by the Schrödinger equation is deterministic for closed systems. Given the initial state $|\psi(t_0)\rangle$ and the Hamiltonian operator H, the state $|\psi(t)\rangle$ at any future time t can be precisely determined.

- However, if the system is not completely isolated and is subject to external influences or measurements, then its time-evolution would be described by a more general framework called quantum mechanics with open systems or quantum mechanics with measurements. In these cases, the time-evolution may involve non-unitary processes and probabilistic outcomes.

2.2.3 Composite Systems

The Composition of Systems Postulate is a fundamental principle in quantum mechanics that describes how the state space of a composite quantum system is formed when multiple subsystems are combined. It tells us how to mathematically represent the joint state of a system composed of individual subsystems.

Composition of Systems Postulate

- The postulate states that When two physical systems are treated as one combined system, the state space of the combined physical system is the tensor product space $H_1 \otimes H_2$ of the state spaces H_1, H_2 of the component subsystems. If the first system is in the state $|\psi_1\rangle$ (pronounced "ket psi one") and the second system in the state $|\psi_2\rangle$. then the state of the combined system is $|\psi_1\rangle \otimes |\psi_2\rangle$ (pronounced "ket psi one tensor product ket psi two").
- This tensor product is not just a simple combination of vectors; it creates a new vector in the combined space, considering all possible ways the two systems can interact and be correlated. This allows us to describe how the two systems influence each other and share information in a very powerful way within the framework of quantum mechanics.
  - We can extend this concept to n distinct subsystems by using the Composition of Systems Postulate. If we have n subsystems, each with a dimension of 2 (for simplicity), the state space of the entire system is the tensor product of the state spaces of the individual subsystems.
  - It's important to note that the state of a combined system cannot always be written as a simple product of states, like $|\psi_1\rangle \otimes |\psi_2\rangle$. If the subsystems are independent and kept isolated, we can write the state in that form.
  - However, if the subsystems can interact, then the combined system may become entangled and its state cannot be expressed in a simple product form.
  - When we apply an operation like the not gate (X gate) to a specific qubit in a composite system, we can represent it as a tensor product of the gate with the identity operator applied to the other qubits.
  - For example, applying the X gate to the first qubit in a 2-qubit system is represented by the operator $X \otimes I$. Thus the 2-qubit input $|\psi_1\rangle \otimes |\psi_2\rangle$ gets mapped to $X|\psi_1\rangle \otimes |\psi_2\rangle = (X \otimes I)\ (|\psi_1\rangle \otimes |\psi_2\rangle)$.
  - That is, the linear operator describing this operation on the composite system is $X \otimes I$ which has the matrix representation

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

- Although it acts on the composite 2-qubit system, since it acts non-trivially on only one of the qubits, a gate such as this one is really a 1-qubit gate.
- Similarly, there are certain gates, like the quantum controlled-not (cnot) gate, that cannot be expressed as a simple tensor product of two 1-qubit gates. The cnot gate flips the state of the second qubit if the first qubit is in state $|1\rangle$ and does nothing otherwise.
- In summary, the Composition of Systems Postulate tells us that we can use the tensor product to describe composite quantum mechanical systems, allowing us to understand how they behave and interact with each other, especially when entanglement is involved.

2.2.4 Mixed States and General Quantum Operations

In quantum mechanics, the concept of mixed states and general quantum operations is essential for describing the behavior of open quantum systems and situations where we have incomplete information about the quantum state.

Mixed States:

- In quantum mechanics, a pure state is a state that can be represented by a single ket vector in the Hilbert space and has a definite and unique quantum state. For example, a pure state could be represented as $|\psi\rangle$.
- On the other hand, a mixed state is a statistical ensemble of pure states. It represents a situation where the quantum system is in one of several possible pure states, each with a certain probability. In other words, the system is in a superposition of pure states, but we don't know which specific state it is in until a measurement is made.
- Mathematically, a mixed state is described by a density operator, often denoted as ρ. The density operator is a positive semidefinite, Hermitian operator that has a trace equal to 1. For a quantum system with dimension N (i.e., N-dimensional Hilbert space), the density operator ρ can be expressed as an $N \times N$ matrix.
- A density operator ρ for a mixed state can be written as a convex combination of projectors (which represent pure states). For example, for a two-level quantum system (qubit), the density operator ρ can be written as:

$$\rho = \rho|\psi_1\rangle\langle\psi_1| + (1-\rho)|\psi_2\rangle\langle\psi_2|$$

where $|\psi_1\rangle$ and $|\psi_2\rangle$ are two orthogonal pure states and ρ is the probability of finding the system in state $|\psi_1\rangle$.

Partial Trace

- Partial trace is a mathematical operation used when dealing with composite quantum systems. It allows us to obtain the reduced density operator for one of the subsystems by tracing out (or summing over) the other subsystems.
- Consider a quantum system that consists of two or more subsystems, each associated with its own Hilbert space. The composite state of the entire system is represented by a ket vector in the tensor product space of the individual subsystems. For example, if we have two subsystems A and B and their state vectors are $|\psi\rangle_A$ and $|\varphi\rangle_B$, respectively, then the composite state is $|\psi\rangle_A \otimes |\varphi\rangle_B$.
- Now, suppose we are interested in studying only the properties of subsystem A, while disregarding the details of subsystem B. To do this, we perform a partial trace over subsystem B. The partial trace operation results in a density operator for subsystem A, representing the reduced state of A.

  Mathematically, the partial trace operation for subsystem B can be written as follows:

$$\rho_A = T_{r_B}(|\psi\rangle\langle\psi|)$$

  where:

  $|\psi\rangle$ is the composite state of the entire system in the tensor product space of subsystems A and B,

  $\langle\psi|$ is the conjugate transpose of $|\psi\rangle$,

  T_{r_B} denotes the trace operation over subsystem B,

  ρ_A is the reduced density operator of subsystem A.

- The reduced density operator ρ_A describes the state of subsystem A, taking into account the information about subsystem B that is averaged out by the partial trace.

General Quantum Operations:

- In quantum mechanics, we often encounter situations where a quantum system interacts with its environment undergoes measurements or experiences noise and decoherence. These processes are described by general quantum operations, which are also known as quantum channels.
- A quantum operation, represented by the operator Φ, describes how the quantum state of a system transforms under the influence of external influences or measurements.

- Unlike unitary operators (which describe reversible and deterministic transformations), general quantum operations can be non-unitary and may cause irreversible changes to the quantum state.
- These operations are often used to model the behavior of open quantum systems, where information or energy is exchanged with the surrounding environment.
- Mathematically, a general quantum operation Φ can be represented as a completely positive, trace-preserving (CPTP) map. A CPTP map ensures that the probabilities of outcomes are preserved and it allows for a consistent description of quantum processes.
- These operations can be described using either Kraus operators, Choi matrices or superoperators, depending on the context.
- General quantum operations play a crucial role in quantum information processing, quantum error correction and quantum communication. They are fundamental for understanding the behavior of quantum systems in realistic scenarios, where interactions with the external world and uncertainty about the initial state are common.

2.2.5 Universal Sets of Quantum Gates

- In quantum computing, a universal set of quantum gates refers to a specific set of quantum gates that can be used together to approximate any quantum computation to arbitrary precision. This means that with only the gates from a universal set, we can perform any quantum algorithm or operation that is possible on a quantum computer, even though it may require a large number of gates and qubits.
- Suppose we approximate a desired unitary transformation U by some other unitary transformation V. The error in the approximation is defined to be

$$E(U, V) = = \max_{|\psi\rangle} \| (U - V)|\psi\rangle\|$$

- An operator U can be 'approximated to arbitrary accuracy', we mean that if we are given any error tolerance $\varepsilon > 0$, we can implement some unitary V such that $E(U, V) < \varepsilon$.
- A set of gates is said to be universal if for any integer $n \geq 1$, any n-qubit unitary operator can be approximated to arbitrary accuracy by a quantum circuit using only gates from that set.
- A 2-qubit gate is said to be an entangling gate if for some input product state $|\phi\rangle|\phi\rangle$ the output of the gate is not a product state (i.e. the output qubits are entangled)
- A set composed of any 2-qubit entangling gate, together with all 1-qubit gates, is universal.
- A set of gates is said to be universal for 1-qubit gates if any 1-qubit unitary gate can be approximated to arbitrary accuracy by a quantum circuit using only gates from that set.
- A universal set of quantum gates typically consists of one or more single-qubit gates and at least one two-qubit gate. The most commonly used universal set of quantum gates is the following:

Single-Qubit Gates:

- **Hadamard Gate (H):** It transforms the basis states $|0\rangle$ and $|1\rangle$ into superpositions of themselves. H is defined as that gate mapping the computational basis states as follows:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$
$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

The Hadamard gate is represented by the following matrix:

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- One useful property of the Hadamard gate is that it is self-inverse, meaning $H = H^{-1}$ and so

$$H(|0\rangle + |1\rangle)/\sqrt{2} = |0\rangle$$
$$H(|0\rangle - |1\rangle)/\sqrt{2} = |1\rangle$$

- **π/8 Phase Gate (T):** It introduces a phase of π/4 to the state |1⟩. T, which acts on the computational basis state follows:

$$T|0\rangle = |0\rangle$$

$$T|1\rangle = e^{i\frac{\pi}{4}}|1\rangle$$

The π/8 – phase gate has the following matrix representation:

$$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{bmatrix}$$

T is equivalent to,

$$\begin{bmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{bmatrix}$$

Two-Qubit Gate:

- **Controlled-NOT (CNOT) Gate:** It flips the target qubit (second qubit) if and only if the control qubit *(first qubit) is in* state |1⟩.

$$\text{Controlled NOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- Using these gates, any unitary operation on a quantum state can be approximated to any desired precision by composing a sequence of gates. This property is known as the universality of the gate set.
- It is important to note that although a universal set of gates is sufficient to perform any quantum computation, the efficiency and complexity of the computation may vary depending on the specific set of gates chosen. Researchers aim to find efficient universal gate sets that can implement a wide range of quantum algorithms with as few gates and qubits as possible, as this reduces the resources required for quantum computations.

2.2.6 Quantum Measurement and Quantum Entanglement

- Quantum measurement is a fundamental process in quantum mechanics that allows us to extract information from a quantum system. When we measure a quantum system, we obtain a specific outcome or result associated with a particular observable property of the system. The process of measurement is probabilistic, meaning that the outcome is not determined with certainty but rather determined by the probabilities associated with the quantum state.
- In quantum mechanics, the state of a quantum system is described by a wave function or state vector, often denoted as a "ket" (|ψ⟩). When a measurement is performed on this state, the wave function collapses into one of the possible eigenstates of the observable being measured. The probability of obtaining a specific outcome is related to the squared magnitude of the probability amplitude associated with that eigenstate.

  For example, consider a qubit in a superposition state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. When we measure this qubit in the computational basis (basis states |0⟩ and |1⟩), the probabilities of obtaining |0⟩ and |1⟩ are given by $|\alpha|^2$ and $|\beta|^2$, respectively. After the measurement, the qubit will be in the state |0⟩ with probability $|\alpha|^2$ or in the state |1⟩ with probability $|\beta|^2$.
- It is important to note that measurement causes the collapse of the quantum state and subsequent measurements on the same system may yield different outcomes due to the probabilistic nature of quantum mechanics.
- Quantum measurement is done by having a closed quantum system interact in a controlled way with an external system from which the state of the quantum system under measurement can be recovered.

Quantum Entanglement:

Quantum entanglement is a unique and fascinating phenomenon that arises in certain quantum systems when two or more particles become correlated in such a way that their states are interdependent, regardless of the physical distance between them.

Entangled particles are described by a joint quantum state that cannot be expressed as a simple product of individual states.

For example, consider two entangled qubits, commonly referred to as "Bell states":

$$|\Phi^+\rangle = (1/\sqrt{2})(|00\rangle + |11\rangle)$$

$$|\Phi^-\rangle = (1/\sqrt{2})(|00\rangle - |11\rangle)$$

$$|\psi^+\rangle = (1/\sqrt{2})(|01\rangle + |10\rangle)$$

$$|\psi^-\rangle = (1/\sqrt{2})(|01\rangle - |10\rangle)$$

- If two qubits are in one of these Bell states, measuring the state of one qubit instantaneously determines the state of the other qubit, regardless of the distance between them. This phenomenon is often referred to as "spooky action at a distance" and has been experimentally verified through tests of Bell's inequalities.
- Quantum entanglement is a crucial resource in quantum computing and quantum communication, enabling various applications, such as quantum teleportation and quantum cryptography. It plays a central role in understanding the non-local and probabilistic nature of quantum mechanics.

2.3 THE QUANTUM FOURIER TRANSFORM AND ITS APPLICATIONS

Phase Estimation : Quantum Fourier transform, which is the key ingredient for quantum factoring and many other interesting quantum algorithms.

2.3.1 The Quantum Fourier Transform (QFT)

- One of the best ways to solve problems in mathematics or computer science is to change them into different problems that we already know how to solve. Some of these transformations are so common and appear in many different situations that they are studied on their own.
- Quantum computing has made a fascinating discovery that some of these transformations can be done much faster on a quantum computer than on a regular computer. One such transformation is the discrete Fourier transform.
- Discrete Fourier transform takes as input a vector of complex numbers, $a_0, \ldots, a_{N-1}$ with a fixed parameter N representing the vector's length. It produces the transformed data, a vector of complex numbers $b_0, \ldots, b_{N-1}$, defined by the formula:

$$b_k = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} a_i e^{\frac{2\pi jik}{N}}$$

- The quantum Fourier transform is precisely the same transformation. The quantum Fourier transform on an orthonormal basis $|0\rangle, \ldots, |N-1\rangle$ is defined as a linear operator with the following action on the basis states:

$$|i\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} a_i e^{\frac{2\pi jik}{N}} |K\rangle$$

Or the unitary matrix:

$$U_{QFT} \rightarrow \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} \sum_{k=0}^{N-1} a_i e^{\frac{2\pi jik}{N}} |K\rangle\langle i|$$

- Equivalently, the action on an arbitrary state may be written

$$\sum_{i=0}^{N-1} a_i|i\rangle \rightarrow \sum_{k=0}^{N-1} b_k|k\rangle$$

Where the amplitudes b_k are the discrete Fourier transform of the amplitudes a_i.

- Although not immediately evident from the definition, this transformation is unitary, which means it can implemented as the dynamics for a quantum computer.
- The quantum Fourier transform can be given the following useful product representation:

$$|i_i, \ldots, i_n\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i_0 i_n}|1\rangle)\,(|0\rangle + e^{2\pi i_0 i_{n-1} i_n}|1\rangle) \ldots (|0\rangle + e^{2\pi i_0 i_1 \cdots i_n}|1\rangle)}{2^{\frac{n}{2}}}$$

- This product representation is so useful that you may even wish to consider this to be the definition of the quantum Fourier transform.

Example 1: 1-qubit QFT

Consider how the QFT operator as defined above acts on a single qubitstate $|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$. In this case $x_0 = \alpha$, $x_1 = \beta$ and N = 2 Then,

$$b_0 = \frac{1}{\sqrt{2}}\left(\alpha \exp\left(2\pi i \frac{0\times 0}{2}\right) + \beta \exp\left(2\pi i \frac{1\times 0}{2}\right)\right)$$

$$= \frac{1}{\sqrt{2}}(\alpha + \beta) \text{ and}$$

$$b_1 = \frac{1}{\sqrt{2}}\left(\alpha \exp\left(2\pi i \frac{0\times 1}{2}\right) + \beta \exp\left(2\pi i \frac{1\times 1}{2}\right)\right)$$

$$= \frac{1}{\sqrt{2}}(\alpha - \beta)$$

such that the final result is the state

$$U_{QFT}|\varphi\rangle = \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle$$

This operation is exactly the result of applying the Hadamard operator (H) on the qubit:

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

After applying the H operator to the state $|\varphi\rangle = \alpha|0\rangle + \beta|1\rangle$, the new state is:

$$\frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle \equiv \tilde{\alpha}|0\rangle + \tilde{\beta}|1\rangle$$

transformed data, a vector of complex numbers $y_0, \ldots, y_{N-1}$, defined by

- Initially, this might seem like a fantastic idea, especially considering how crucial the Fourier transform is in various real-world data processing tasks.
- For instance, in computer speech recognition, the first step involves Fourier transforming the digitized sound to recognize phonemes.

2.3.2 Phase Estimation

- The Fourier transform plays a crucial role in a general process called phase estimation, which serves as a building block for many quantum algorithms.
- In this process, we aim to estimate the phase ϕ of a unitary operator U, which has an unknown eigenvalue $e^{(2\pi i\phi}$ associated with an eigenvector $|u\rangle$.

To perform phase estimation, we assume the availability of black boxes (also known as oracles) capable of preparing the state $|u\rangle$ and performing controlled-$U^{(2^j)}$ operations, where j is a suitable non-negative integer.

These black boxes serve as essential tools for the phase estimation procedure but are not complete quantum algorithms by themselves. Instead, phase estimation acts as a "subroutine" or "module" that, when combined with other subroutines, can accomplish interesting computational tasks.

- The quantum phase estimation procedure utilizes two registers. The first register contains t qubits initially in the state $|0\rangle$.
- The choice of t depends on two factors: the number of digits of accuracy required for the estimation of ϕ and the desired probability of success for the phase estimation procedure. The value of t naturally emerges from the subsequent analysis, considering these factors.
- The second register starts in the state $|u\rangle$ and contains enough qubits to store $|u\rangle$. The phase estimation is carried out in two stages.
- First, we apply a specific circuit described in Fig. 2.3. The circuit begins by applying a Hadamard transform to the first register, followed by controlled-U operations on the second register, with U raised to the power of 2^j.
- The final state of the first register is easily seen to be:

$$\frac{1}{2^{\frac{t}{2}}}(|0\rangle + e^{2\pi i 2^{t-1}\varphi}|1\rangle)(|0\rangle + e^{2\pi i 2^{t-2}\varphi}|1\rangle)\dots(|0\rangle + e^{2\pi i 2^{0}\varphi}|1\rangle) = \frac{1}{2^{\frac{t}{2}}}\sum_{k=0}^{2^t-1} e^{2\pi i\varphi k}|k\rangle$$

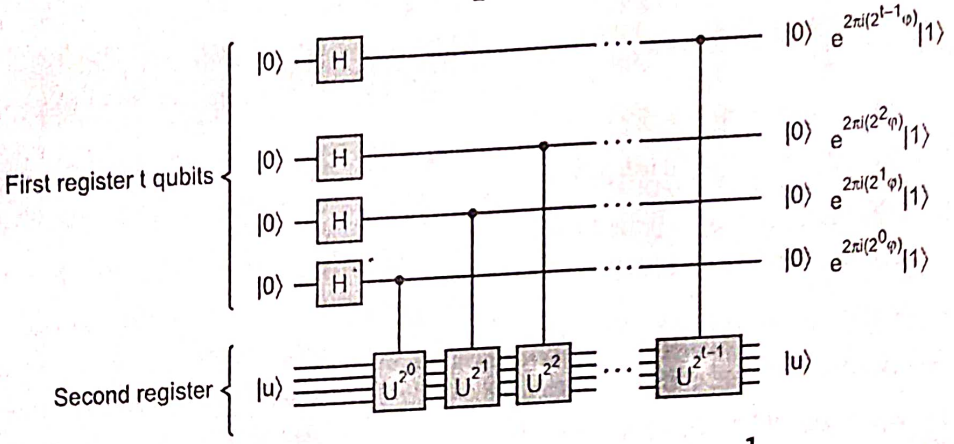

Fig. 2.3: The first stage of the phase estimation procedure. Normalization factors of $\frac{1}{\sqrt{2}}$ have been omitted, on the right

- In the second stage of phase estimation, we perform the inverse quantum Fourier transform on the first register. The inverse quantum Fourier transform can be accomplished in $\Theta(t^2)$ steps.
- In the third and final stage of phase estimation, we read out the state of the first register through a measurement in the computational basis. This measurement provides us with a pretty good estimate of the phase ϕ.
- The overall algorithm is schematically shown in Fig. 2.4.

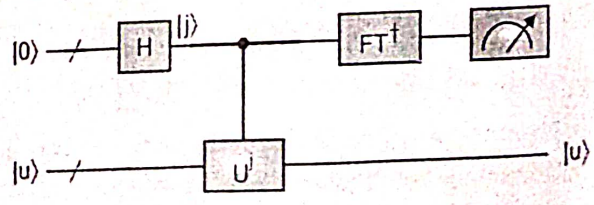

Fig. 2.4: Schematic of the overall phase estimation procedure

- The top t qubits (the '/' denotes a bundle of wires, as usual) are the first register and the bottom qubits are the seco register, numbering as many as required to perform U.|u⟩ is an eigenstate of U with eigenvalue $e^{2\pi i\varphi}$.
- The putput of the measurement is an approximation to φ accurate to $t - \left[\log\left(2 + \frac{1}{2e}\right)\right]$ bits, with probability o success at least $1 - \epsilon$.

2.3.3 Order-finding and Factoring

- "Order-finding" and "factoring" are two important problems in the field of quantum computing that have significant implications for cryptography and number theory. They are closely related and solving one of these problems efficiently can lead to the solution of the other.
  - ➢ **Order-Finding:** The order-finding problem is a number-theoretic problem that arises in the context of modular exponentiation. Given a positive integer "a" and another positive integer "N" that is coprime to "a" (that is, they have no common factors except for 1), the order-finding problem seeks to find the smallest positive integer "r" such that:

$$a^{\wedge r} \equiv 1 \pmod N$$

where "^" denotes exponentiation and "mod N" means taking the remainder when dividing by N.

The order "r" is a critical value in number theory and it has applications in various fields, including cryptography. In particular, the order-finding problem is a key component of Shor's algorithm, one of the most famous quantum algorithms.

The order-finding algorithm is summarized below.

Order-Finding Algorithm

1. Choose an integer n so that $2^n \geq 2r^2$.
   The value n = [2 logN] will suffice.
2. Initialize an n-qubit register to $|0\rangle^{\otimes n}$.
   Call this the control register.
3. Initialize an n-qubit register to $|1\rangle = |00 \ldots 01\rangle$.
   Call this the target register.
4. Apply the QFT to the control register.
5. Apply c- U^x_a control and target registers.
6. Apply the QFT^{-1} to the control register.
7. Measure the control register to obtain an estimate $x_1/2^n$ of a random integer multiple of 1/r.
8. Use the continued fractions algorithm to obtain integers c_1 and r_1 such that $\left|\frac{x_1}{2_n} - \frac{C_1}{r_1}\right| \leq \frac{1}{2^{\frac{n-1}{2}}}$.
   If no such pair of integers is found, output 'FAIL'.
9. Repeat Steps 1–7 to obtain another integer x_2 and a pair of integers c_2 and r_2 such that $\left|\frac{x_2}{2^n} - \frac{C_2}{r_2}\right| \leq \frac{1}{2^{\frac{n-1}{2}}}$. If no such pair of integers is found, output 'FAIL'.
10. Compute $r = LCM(r_1, r_2)$. Compute a^r mod N.
11. If a^r mod N = 1, then output r. Otherwise, output 'FAIL'.

- **Factoring:** Factoring is the problem of decomposing a composite number "N" into its prime factors. For example, if N = 15, the goal is to find its prime factors, which are 3 and 5, since 3 × 5 = 15. Factoring large numbers is a challenging problem, especially for classical computers, as the best-known algorithms have exponential time complexity.

  However, Shor's algorithm, a quantum algorithm developed by mathematician Peter Shor, can efficiently factor large numbers in polynomial time. The central step of Shor's algorithm involves the order-finding problem, where finding the order of a random integer modulo N helps to efficiently factor N.

- **Significance for Cryptography:** The factorization problem is essential for cryptography, particularly for public-key cryptography schemes based on the difficulty of factoring large numbers. Popular cryptographic algorithms, such as RSA, rely on the assumption that factoring large numbers is computationally infeasible for classical computers.

  Quantum computers, if large enough and error-tolerant, could efficiently solve the order-finding and factoring problems using Shor's algorithm. This poses a significant threat to certain cryptographic schemes, as quantum computers could break widely-used public-key cryptographic systems like RSA, rendering them insecure.

3.4 General Applications of the Quantum Fourier Transform-Period-Finding

The Quantum Fourier Transform (QFT) has several general applications in quantum computing and one of the most prominent applications is in period-finding. The period-finding problem is a fundamental computational problem with applications in cryptography, number theory and quantum algorithms, particularly in the context of Shor's algorithm for integer factorization.

Period-finding involves finding the period of a periodic function or sequence. In the context of Shor's algorithm, the period-finding problem is used to efficiently factorize large composite numbers. Given a positive integer "N" and another integer "a" that is coprime to "N" (that is, they have no common factors except for 1), the goal is to find the smallest positive integer "r" such that:

$$a^{\wedge r} \equiv 1 \pmod{N}$$

where "^" denotes exponentiation and "mod N" means taking the remainder when dividing by N.

The quantum Fourier transform plays a crucial role in Shor's algorithm for period-finding as follows:

- **Quantum Superposition:** The quantum Fourier transform allows us to create superpositions of states representing different possible periods. By applying the QFT to a quantum state that represents all possible candidates for "r," the algorithm considers all potential periods simultaneously. This property is known as quantum parallelism and it is the key to the efficiency of Shor's algorithm.
- **Quantum Interference:** The QFT introduces quantum interference, which is a phenomenon that arises when different states in the superposition interfere constructively or destructively. The interference enhances the probability of measuring the correct period "r" and suppresses the probabilities of measuring incorrect periods.
- **Quantum Amplitude Amplification:** Shor's algorithm employs quantum amplitude amplification to further enhance the probability of obtaining the correct period in a relatively small number of iterations. By iteratively applying a series of quantum gates to amplify the amplitude of the correct period state, the algorithm significantly improves its efficiency.

As a result of these quantum properties, Shor's algorithm using the quantum Fourier transform can efficiently find the period "r" in polynomial time. Once the period is obtained, it can be used to factorize the composite number "N."

Period-finding using the quantum Fourier transform is a powerful application of quantum computing with potential implications for cryptography and number theory. It shows the advantage of quantum algorithms over classical algorithms for solving specific computational problems and is a prime example of the potential of quantum computing for speeding up calculations in certain domains.

2.3.5 General Applications of the Quantum Fourier Transform- Discrete Logarithms

- Discrete logarithms are an essential mathematical problem in number theory and cryptography. The discrete logar problem is as follows: Given a prime number "p," a base "g," and an integer "y," find the exponent "x" such that:

$$g^{\wedge x} \equiv y \pmod{p}$$

where "^" denotes exponentiation and "mod p" means taking the remainder when dividing by the prime number "p

- The discrete logarithm problem is considered computationally hard for classical computers, especially when the pri "p" is large and is the basis for certain cryptographic schemes, such as Diffie-Hellman key exchange and digi signature algorithms like DSA.
- Here's how the QFT is applied in these algorithms:
  - ➢ **Quantum Superposition:** The QFT allows us to create superpositions of states representing different possibl values of the discrete logarithm "x." By applying the QFT to a quantum state that represents all possible candidate for "x," the algorithm considers all potential values simultaneously. This property is known as quantum parallelism and it is the key to the efficiency of quantum algorithms for discrete logarithms.
  - ➢ **Quantum Interference:** The QFT introduces quantum interference, where different states in the superposition interfere constructively or destructively. The interference enhances the probability of measuring the correct value of the discrete logarithm "x" and suppresses the probabilities of measuring incorrect values.
  - ➢ **Quantum Amplitude Amplification:** Quantum algorithms for discrete logarithms use techniques like quantum amplitude amplification to further enhance the probability of obtaining the correct value of "x" in a relatively small number of iterations. This amplification process significantly improves the efficiency of the algorithm.
- The combination of quantum parallelism, quantum interference and quantum amplitude amplification allows quantum algorithms to solve the discrete logarithm problem more efficiently than any known classical algorithm. This has significant implications for cryptography, as the security of many cryptographic schemes relies on the difficulty of computing discrete logarithms.

  For example, the efficiency of a quantum algorithm for discrete logarithms could threaten the security of widely used cryptographic protocols like Diffie-Hellman key exchange and digital signatures based on the discrete logarithm problem. This has prompted research into post-quantum cryptography, which seeks to develop cryptographic schemes that are resistant to attacks from both classical and quantum computers.
- In summary, the quantum Fourier transform enables efficient quantum algorithms for solving the discrete logarithm problem, which has implications for the security of cryptographic systems and demonstrates the power of quantum computing in tackling complex number-theoretic problems.

2.3.6 General Applications of the Quantum Fourier Transform-The Hidden Subgroup Problem

The hidden subgroup problem (HSP) is a computational problem in the field of quantum computing and is a generalization of several other problems, including period-finding and discrete logarithms. It can be understood as follows:

- **Given a group G and a function f:** $G \rightarrow X$, where X is a set of labels, the hidden subgroup problem seeks to find a subgroup H of G that is invariant under the function f. In other words, for every pair of elements g and h in H, the function f maps them to the same label:

$$f(g) = f(h) \text{ for all } g, h \text{ in } H.$$

The goal of the hidden subgroup problem is to identify this hidden subgroup H, which is generally hidden in the structure of G and not directly accessible through the function f.

- General applications of the quantum Fourier transform (QFT) to the hidden subgroup problem: The QFT plays a centra role in quantum algorithms designed to solve the hidden subgroup problem efficiently.

Here's how the QFT is applied in these algorithms:

- **Quantum Superposition:** The QFT allows us to create superpositions of states representing different possible subgroups H of the group G. By applying the QFT to a quantum state that represents all possible candidates for the hidden subgroup, the algorithm considers all potential subgroups simultaneously. This property is known as quantum parallelism and it is the key advantage of quantum algorithms over classical algorithms for the hidden subgroup problem.
- **Quantum Interference:** The QFT introduces quantum interference, where different states in the superposition interfere constructively or destructively. The interference enhances the probability of measuring the correct subgroup H and suppresses the probabilities of measuring incorrect subgroups.
- **Quantum Amplitude Amplification:** Quantum algorithms for the hidden subgroup problem use techniques like quantum amplitude amplification to further enhance the probability of obtaining the correct subgroup H in a relatively small number of iterations. This amplification process significantly improves the efficiency of the algorithm.

By exploiting the quantum properties of superposition, interference and amplitude amplification, quantum algorithms based on the QFT can efficiently solve the hidden subgroup problem in polynomial time. This has implications for various applications, including cryptography, optimization and quantum error correction.

The hidden subgroup problem is a foundational problem in quantum computing and its efficient solution demonstrates the power of quantum algorithms in tackling complex computational problems with potential real-world applications. Additionally, the hidden subgroup problem provides a rich theoretical framework for understanding the capabilities and limitations of quantum computing in comparison to classical computing.

CASE STUDY: FPGA-BASED QUANTUM CIRCUIT EMULATION: A CASE STUDY ON QUANTUM FOURIER TRANSFORM

FPGA-Based Quantum Circuit Emulation is a research area that focuses on using Field-Programmable Gate Arrays (FPGAs) to simulate and emulate quantum circuits. This approach allows researchers and engineers to study and prototype quantum algorithms and circuits efficiently. In this case, you mentioned a specific case study on the Quantum Fourier Transform (QFT).

Here are the key aspects of FPGA-based quantum circuit emulation:

- **Quantum Fourier Transform (QFT):** The Quantum Fourier Transform is a fundamental quantum algorithm used in quantum computing. It plays a crucial role in many quantum algorithms, including Shor's algorithm for integer factorization and quantum phase estimation. QFT is analogous to the classical discrete Fourier transform but operates on quantum states. It's used for quantum state manipulation and information processing.
- **Field-Programmable Gate Arrays (FPGAs):** FPGAs are hardware devices that can be programmed to implement digital circuits. They consist of an array of configurable logic blocks and programmable interconnects. Unlike general-purpose CPUs or GPUs, FPGAs can be customized to perform specific tasks efficiently.
- **FPGA-Based Emulation:** FPGAs are hardware devices that can be programmed to implement specific digital circuits. In the context of quantum computing, FPGA-based emulation involves using FPGAs to simulate and model quantum circuits, allowing researchers to analyze and experiment with quantum algorithms in a controlled and efficient manner.

Advantages of FPGA-Based Emulation:

- **Parallelism:** FPGAs offer inherent parallelism, making them well-suited for parallel processing, which is essential for quantum circuit simulation.

Low Latency: FPGAs can provide low-latency processing, which is important for real-time simulation and hardware acceleration of quantum algorithms.

Energy Efficiency: FPGAs can offer energy-efficient emulation, making them suitable for large-scale quantum simulations.